

How Do Dreams Relate To Consciousness? Why We Dream & The Felt Uncertainty Theory | Mark Solms

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[Music] hey Mark how's it going fine yourself great man great great great great to have you back good to be back although the word round two makes it sound like a boxing match yeah no it's just uh it's just a way to keep track of the conversations nothing serious how you doing are you still in Boston mark no I I got back um two days ago okay how was the how was the tri it was actually a slightly nightmarish flight to Boston I got um diverted eventually to New York okay and um and then it was of a shambles but apart from that part of it I mean I ended up losing almost a whole day but the once I did get there scientifically it was just fantastic I really enjoyed it I know I saw some of you did a presentation at tough um yeah it seemed like a really great time I did a presentation which I was happy with I was very happy with how it was received by some people who I respect you know so it's it's it's it's good um to to get good feedback from Minds that you admire and then there was also the added dimension of Mike's lab I mean I know about his work but to actually visit his lab it's just quite extraordinary I'm actually to touch on that at some point because I wrote a note here about the collaborations you guys are doing friston Fields Le I mean there's so much happening here and it's it's so exciting from the outside I asked Mike about this about a week ago and it just seems incredible he left me with a nice question for you right at the very end what is the meaning of life you know I tweeted this to you Mike also he just wants to know he thinks you actually might provide a very valuable answer to this cuz he thinks that your background in neuropsychology will help enhance this answer so give us that well um actually I'd forgotten that you tweeted that I wish i' I wish I'd prepared an answer because that's not the kind of question one likes to receive what is the meaning of life especially when you're expecting a serious and and and uh and satisfying answer so um what what I would say first of all is we must distinguish between uh the the the sort of uh purpose of life and the meaning of life the purpose of life is very easy to discern it's to keep on going being alive you know it's to to

survive and to reproduce I'm afraid is the purpose of life described biologically because of course life is a biological thing and so when you speak of what is it for it's just for keeping it going that's all it's for um but that doesn't mean that's the meaning of life you know so I I would draw a distinction between us as biological organisms uh which are just struggling to um resist the second law of Thermodynamics and and and keep going or resisted for as long as we can and then we lose you know the I would like to distinguish between that and the the the being of an organism you know our subjective existence that's where I think the word meaning uh is appropriate it's like okay so I'm here what what am I going to you know what should what should I do with this other than just you know Carry On from a biological point of view existing as a as a living thing what is the meaning of all of this um and my um my emphasis of course is on feelings um as you know I think subjectivity is fundamentally um constituted by feeling we are feeling beings that's what being is to be to to for the you know that phrase there's an organism that that there's something it is like to be that organism it it is a feeling organism um now our feelings uh again as you know because I've spoken to you about before they are just the way in which we become aware of our needs so so you could look at that biologically you could say well the reason why for example the reason why pleasure is attached to sexuality is because that's how our species reproduces and so from a purely biological point of view sex is pleasurable uh because that's that's got to do with that purpose side of things the meaning side of it I mean why do we actually have sex as subject objects you know as as feeling beings it's it's not because we're trying to do our biological Duty you know not at all um we we we in fact most of the time that we have sex we are hoping not to reproduce so it's there really is it's not a small distinction it's it's it's a very different thing um what we're doing in terms of in terms of what motivates us as sentient beings as subjects um has has to do with so it's for that reason for me it has been a very important part of the meaning of my life is to try and understand you know what is this what is feeling uh what what what is it what is it all about um and to try and understand it I think that that's been a very worthwhile quest uh to try to understand what is it to be and what is feeling and um the the I person can't find a more um a more um worthwhile way of spending my brief period of biological existence trying to understand what what it is what it is all about um and uh I'm I'm perfectly happy with the meaning of my life from from that point of view now I used sex as a as an example because it's so easy to see the biological versus the subjective but it applies to all of our needs you know what we what we're living is it's like let me take something even more praic you know we don't uh we we eat chocolate and sweets and stuff you know because

they taste nice the the reason they taste nice is because they've got high energy content that's what makes them taste nice but we're not thinking oh you know the what I really need to do is get high energy supplies what we're doing is saying I really like this you know so um the the other main thing I've done in my life is is is train in and practice psychoanalysis and that's really uh you could easily say uh that it's not the only way of going about it but it is one of the it's a it's like like the many Eastern traditions where like meditative traditions and so on in fact I met a very senior Buddhist when was in Boston now and so I spoke quite a bit um or interacted quite a bit with with um with that way of approaching the same task which is you know really trying to properly confront what is it to be what is it to be mortal what is what are these feelings really for what do I feel about them what are what what are my values what do I want to do in relation to my felt needs which include needs to uh not only survive and reproduce and eat uh but but you know to have fun uh Play It's terribly important you know to recognize that has been a a real Discovery in my life and and to recognize it as a you know as an experien of of the need to play the need to have fun to rec so the more I've understood about you know how minds work and how my mind works the the better I'm able to enjoy my mind uh and and recognize that it's inevitable that there's no such thing as a conflict free existence you know all of our different needs are they they need to be balanced with each other uh and you kind of you become more sanguin about it all and and allow yourself to appreciate you know what what it is what it is to to and foregrounding feelings has been a a very important um not not only because of my own work in relation to the to the neuroscience of feeling but also you know in relation to psychoanalysis getting to know your own feelings face up to them face up to your conflicts Ponder how are you going to deal with them recognize you're never going to be Exempted from conflicts and so on so I suppose um what is that um there's a phrase the examined life yesam life is not worth living yeah and uh and also the old you know know thyself I think that's the that's that's that is the that is the meaning that you end up that I end up realizing that's the best you can do with with life is to is to is to wonder at it to ponder it to try to understand it to reflect upon it to be fully alive in the sense of being alive you know not not not biological life but but experiential life um to to to to to to to ponder it to reflect on it to study it to to wonder at it to appreciate it I think I think we're similar in that case and this podcast obviously embodies that that Curiosity that that sort of this quest for us to try and Conquer his mind when I spoke to Mike this that question came at the very end and I remember it's because apparently Mike gets a lot of emails lately regarding the meaning of life and and he's not a psychologist so he doesn't have this background he's not a

philosopher so for him he gets caught in a between a rock and a hot place because he's obviously not an expert in those fields so he said he'd rather just shift that onto you I'm not surprised he gets uh emails like that because his work I mean it it really is um pretty fundamental in terms of you know what what are what is an organism and and and and you know what is what is life right at the most basic level of single cells and so on it's it's fantastically interesting what he's doing it's pretty cool and I know that you guys you were in Boston you went you visited Mike's lab um first I don't know if for all of those curious do you want to tell us about that experience it was an extraordinary experience I know about his work um but knowing about something and experiencing it is you know are two different things and that's like of your work pretty much you actually got to feel it yeah so um the the the the main species that he's studying uh he studies various different organisms but these little worms um that are the the first thing that's astonishing about them uh and to actually encounter one such worm um is is that they're Immortal I mean literally Immortal you know they they've been around for millions and millions of years they don't die they just subdivide um and so there's no mixing of genomes you know it's the same genome carrying on and on and on and on and um so in some very real sense you know you're looking at a many millions of years old creature and what's more it's looking back at you it's got these big eyes so there's this encounter with with with with an immortal being um they they um then the the fact that they subdivide um you know is what in what led Mike to to study the mechanisms of morphogenesis how does an organism know what it's meant to become you know um and there how at the most fundamental mechanistic level does that happen then you can chop these creatures in half and the the head grows a tail and the tail grows a head um and so you know having studied what those mechanisms are um he started to fiddle about with them because if you really do understand the mechanism in fact the one way in which you can convince yourself and your colleagues that you do understand the mechanism is to is to make an intervention in it and predict what the consequences will be and if you can predictably control that mechanism then you really do understand it and so as you no doubt know he made a two-headed worm you know by manipulating the the the these bioelectrical fields that are what guide the mor for Genesis so I also met a two-headed worm which was which was something to behold you know I mean it's just really extraordinary and um and then when those worms subdivide they subdivide into two-headed worms so he sort of created a new species yeah I think I think those are the ones he calls xenobots eh I'm not sure if that is the xenobots are yeah say he they various you know like he takes a tadpole and and shifts where its eyes going

to be and they end up with this frog that seizes out of its house and but I mean all these things could be um they they they could be perceived as as ghoulish or or or even as in you know in all sorts of ways inappropriate so it's important to remember that ultimately what it's all about uh is you know to what extent can we translate this basic knowledge from the lab to the clinic and it has it has profound potential implications for how we treat amputees um how we deal with um you know situations where currently one has to do organ transplants uh the possibility of being able to regenerate a diseased organ rather than to transplant it and so on I mean the the possibilities are amazing but then in terms of you said um you know what are we doing in terms of collaborations the the the main overlap it's it's the main overlap because actually there are many ways in which our our um interests and and research uh intersect but the main overlap is um I believe and there's a long story that attaches to that so I'll just come to the conclusion you know I believe for for lots of very good scientific reasons um that the fundamental form of Consciousness is feeling and feelings enable us to know how well or badly we're doing in terms of remaining within our our our viable bounds so moving out of homeostasis feels bad moving back towards homeostasis feels good and um the the what this does that's that's um that's an adaptation Beyond mere homeostasis because many homeostatic mechanisms are purely autonomic you there's no feeling involved like blood pressure regulation and so on um by being able to feel homeostasis feel how well or badly you're doing in terms of your attempts to to to maintain your biological balance enables you to change your mind uh this is not working so before it's too late you you you you you alter um your your your behavior you change your policy and um so this why that's so important is it enables a creature to survive in an uncertain unpredicted environment um so you can come up with novel solutions to Life's Problems by feeling your way through the problem you're not you're not limited to to a single stereotyped response when you find yourself in a situation for which you do not have an innate uh uh uh a um phenetically given response you can feel your way through the situation and come up with a novel response that is what I've just described is the very basis of voluntary Behavior you know which is no small thing and then we learn from from the from the experience and so we can we can adapt to uh unpredicted situations and God knows there many of those in life we can we can do them during our own onto genetic development now the crucial thing is that with Mike fiddling about with these uh creatures I mean what the problem of what do you do when you've got two heads is a novel problem it's it's not something that that species was prepared for by Evolution what happens if you move the eye um to an unexpected location you know this this is not something for which there is

any possibility of there being an automatic inborn um solution and so uh I'm very interested in in what way is the the the sorts of things that these um uh uh very much simpler creatures than ourselves the way that they solving these novel problems uh raises questions of you know how far down does feeling Go is is is a creature like this worm feeling its way through a problem because if it is you know then wow I mean I thought um that I had uh set a cut cut among the pigeons when I told my colleagues I don't believe that cortex is the the the the organ of Consciousness I think that you know the the reticulate activating system and per aqueductal gray you know these structures that we share with all vertebrates so for example with fishes you know um I thought this was Radical but if it goes down even even more than that you know then the implications are just you know gigantic um so that's that's where my work overlaps with with Mike's um and that's what the the the collaborations and there's also Chris Fields I don't know if you know him and and Richard Watson you know so we we've all been in fact Mike is a great um he he he he us he gets the right people together yes um it seems like collaboration yeah he I asked him for a name you could just give me like a a nice name of this group he said he called it the field of diverse intelligence are you happy with that I can't remember what he he he gave the his lab was called something like the center for possible selves or something like that yeah know this is very C in that regard but I was I was I was just it's so fascinating to see all these great minds come together CU then I told him it's almost like the Avengers of mind getting together trying to solve the hard problem it's pretty cool so from our perspective we just want to say thanks tell me so sorry continue I I think that we are living through a sort of a golden age um if not of Neuroscience uh um as a whole certainly Consciousness studies um I I think we really are at at a at a very interesting and important time in the in the history of of that field yeah it it seems to be just escalating so much more and and so much faster that hopefully we're around to see what happens next me it's kind of it's kind of disappointing knowing that we're we're at this Forefront and you never know if you're just going to be alive long enough to see it do you ever get that sort of anticipation anxiety well um it's not the way I experienced that very issue that you're discussing uh it it's more from the other end the way I experience it it is that um when you go against the grain um you know when you say something like I just said now you know I believe that a child who's got no cortex uh that this is this is a conscious being that that there's something it's like to be such a child it it's it it really doesn't win you um a universal approval you know people start saying you know well if if you go against the compact majority view in a science this your science your science defends itself yes um and so what I'm leading up to

saying is that um I have found more courage than I might otherwise have um have had um I'm willing to take risks and to say well you know I know that this is going to um put many noses out of joint I know many people are going to think I'm a Nutter you know that I've lost my way or whatever and I'm I'm I'm less anxious about that than I otherwise would have been because I know you know you only live once and uh as you can see I'm not a kid you know so I I'm just going to throw everything I have at it and try and try and make as much progress as I can uh regardless of the um not regardless of all consequences because I am very concerned about the ethical consequences of some of the issues we're dealing with uh so it's not it's not I don't care about any of the consequences it's that I don't care about the consequences for my reputation um that's what I mean yeah so you've lost that that that that worry about the trivial stuff you're more focused on the bigger picture yeah well I it's because I realize it is the bigger picture you know the there there's there's times uh in my life earlier in my life when I have been very thrown by the fact that um I mean just to take a very earlier in my career um when I made some discoveries that made me realize that what my entire field believed about the brain mechanism of dreaming was wrong you know that that that dreams are not generated by um the part of the brain that generates REM sleep that REM sleep and dreams are not the same thing you know um it it was really a hell of a a hell of a thing to have all of your seniors you know all of the the great authorities in the field uh all just just uh be being um um you know very hostile because uh you were you know you you're spoiling the party you know we you're going against the status quo nobody likes that nobody likes that that complete but I have a note on that so the exact point you're talking about it I've written down some notes so I think we'll touch on that soon let's start at the beginning Mark so the the the whole focus of this conversation for everyone watching or listening is on dreams and the hard problem of Consciousness it's a paper I'll put a link in uh in the description and uh very great paper as always um read many of your papers in the past you even know that I used used one of your papers in Frist for my own dissertation back when I was writing it so um always a pleasure to read your work Mark and yeah I'm looking forward to this first do you want to start with how dreams can assist us with this approach to the hard problem of Consciousness um well look I I have a great many things to say about dreams in general but I'm I'm taking uh your question uh in the narrow way that I that I think I heard it you're saying I think before you do that because I see my next question is probably will address what you want to tell me we spent 20% of all of our sleep dreaming tell me why well that's the interesting thing it's an embarrassment to science Nobody Knows Why um the REM sleep was discovered in the early

1950s um and its high correlation with dreams uh which is a sort of 90% correlation you know um was discovered in the mid 1950s too and um from then onwards they were taken to be one and the same thing dreaming was uh construed as the subjective experience of REM sleep REM sleep was construed as the objective physiological manifestation of of the of the of the of the dream so um everything that we learned in the subsequent decades um about what REM sleep is for because of course um it's a hell of a lot easier to study the mechanisms of something objective than of something subjective so once that correlation had been established the dreams were thrown out as uh like Slippery Fish the they were discarded uh and we and we clung to this much more tractable problem of what is REM sleep what does it do uh and that that's those are easy problems you know because you can manipulate it causally it's very easy and um we we were therefore for decades um we were studying the biological function of REM sleep on the assumption that in so doing we were studying the biological function of dreams but they are doubly dissociable phenomena you can have dreams without REM sleep not only outside of REM sleep but you can have dreams if you are incapable of generating REM sleep due to brain lesion and conversely you can have R sleep without any dreams if you have a lesion to the part of the brain that generates dreams REM is preserved damage to the part of the brain that generates REM dreams are preserved so they are just simply not the same thing therefore everything that we learned about the biological function of REM sleep thinking that we were learning about the biological function of Dreams is um not wasted work because we learned a lot about the biological function of REM sleep but it's wasted from the point of view of what did it teach us about dreams it taught us next to nothing about the function of Dreams um so uh I'll tell you what now only in recent years um have uh have I turned my attention to that question and used appropriate methods because I'm not conflating those two things with each other um I did a pilot study um just before Co um and uh and then I was of course um unable to conclude the study uh because of Co so I've only now couple of years later started the bigger study so I can only tell you uh what we found so far in our pilot study um in order for me to tell you that I need to just take a step backwards and tell you that there are two parts of the brain uh which are essential for dreaming I mean of course the whole brain in a way is essential but there two parts that are really The crucial mechanisms the one which is not surprising is our our perceptual cortex uh the posterior cortex especially visual cortex it's not surprising because what are dreams if not perceptual phenomena and especially visual perceptual phenomena if those if that cortical region is damaged you can't dream because you can't generate the imagery that's NE you know the actual the actual hallucination can't be

generated um but the other part is much more interesting uh it's called the mesocortical mesolimbic dopamine circuit uh and U this is a powerful positive motivational Circuit of the mammal brain uh uh if you stimulate that circuit electrically or chemically it produces the most uh animated exploratory um reward seeking behaviors um in in in the animal now that circuit is essential for dreams uh uh if that's damaged you also can't dream and so you know that raises all sorts of interesting questions um but the the the the one that I was interested in was how on Earth in the middle of sleep with this um as I said it generates the most powerful exploratory Behavior is it it when you look at the the dreaming brain and you see that switched on you know like a Christmas tree you think how on Earth does this thing just lie there sleeping you know it's it seems like a absolute Paradox so the the hypothesis was that the hallucinated dream it's a kind of virtual exploratory and other kinds of emotional motivated Behavior Uh that that that that happens instead of real Behavior so the way way that the animal um um executes its motive and when I say animal of course I'm including ourselves that that um where this motivational urge or surge goes to is into this imagined activity this hallucinated virtual um behavior of the dream which means that if you can't dream due to damage to the posterior cortex if you can't generate the imagery then you should you should wake up because because the dream is there to to enable you to continue sleeping that was the hypothesis and it's a lovely hypothesis because it's entirely counterintuitive most people would think if you can't dream you'll sleep better but on this hypothesis if you can't dream uh you will your sleep will be your sleep will be disturbed and so in our pilot study what we found was that on steroids uh you know you have never seen such terrible sleep as you get in these patients um who can't dream due to due to posterior cortical damage um they have sleep onset insomnia sleep maintenance insomnia they have they have know massively more Awakenings and and and sort of micro Awakenings what we call arousals and microarousals um the Sleep efficiency index is off the charts different Total Sleep time off the charts different so the one thing that um that that although it's only a pilot study so far but the results were so clear-cut I I'm prepared to say it publicly you know the one thing that I believe we can say empirically um we we we have we have demonstrated is that whatever else dreams do they enable us to stay asleep uh uh in the in the uh face of of arousals of things that would otherwise that would otherwise arouse us let's not forget REM sleep is itself an arousal State you know it's it's it's uh when it was first discovered it was called paradoxical sleep that was how come you're able to stay asleep when your brain is so aroused so I'm saying that whatever else dreams do they are guardians of sleep uh they they enable us to to to not have to wake up uh when we are

motivated to to to act uh to to to do something on the basis of um of arousals just are homeostatic deviation so on the basis of homeostatic deviations and Mark in terms of um energy consumption is this is this worth it is this constant need to to experience worth the tradeoff well remember it's in a sense it's the opposite uh of the the question is you know um can we afford not to sleep that's true so you know you the for the whole restitutive function of sleep uh you can't do without that it does there's a hell of a lot of repair work and and um and um housekeeping that is done by the body and the Brain uh and including in when I say housekeeping including memory consolidation processes you know how you process what you've learned during the during the day um there's a lot of very important work that is done um by the body and the Brain while you're asleep and if you if if you don't give yourself enough of that you're a doctor you know how bad it is to not sleep you know it's it's it's a very serious thing to to be deprived of sleep for your health so it's it's can we afford not to sleep and and the energies the energy uh um the work that needs to be done in order to maintain sleep is is well worth the expenditure see it's it's almost intuitive when you think about it I mean we the idea of this I mean because you when you think about the average human we don't really want to not experience it's almost like there's this constant need to continue to have a continuous spectrum of experience that's why a lot of people fear going under anesthesia I mean it's um they know there's almost a switching off of of this this shutdown of the whole Human Experience do you feel like that make sense uh with yeah I I think that um or is it different somehow no no it's I think that what you're saying is is is true but there are also things um which contradict it and um I will I will give you um I think there three things I need to say about that the one is the opposite of what you said which is that we want to we want to retain homeostasis um so so moving out of homeostasis generates negative feelings you know it's like um air hunger or or sleepiness or pain or whatever these are all deviations from homeostasis um and pleasurable feelings are uh returns towards homeostasis but when you're in homeostasis you feel nothing it's satiation so you know I'm hungry that's bad I eat that's good until I've had enough uh and then you know it's it's it's over same with sleep same with thirst same with everything so that's the first thing I wanted to say is that in terms of homeostasis for the most part actually what we're trying to do uh is get out of the zone of feeling and into the zone of automaticity into the zone of satiation um actually feel satiated well you feel satiation satiation as you come to the end of the of the return to homeostasis but then that particular homeostat is off the radar it's you you're not busy thinking I'm I'm not hungry I'm not thirsty I'm I'm not sexually aroused I'm you know I'm not sleepy it's just not

there it's not it's not it's not it's not on the agenda you're feeling uh those things which are going well or badly but once you are uh uh uh at the destination of those regulatory mechanisms then they no longer are are are registered in Consciousness so that as I said that first thing I've said of three things I'm going to say that that's the opposite of what you were saying however um here's the second thing one of those stats is the very one I was talking about when I said this mesocortical mesolimbic dopamine system that has to do with the seeking drive the exploratory foraging uh Drive what what some people call the epistemophilic drive um and and how that homeostat works is is is quite different from the others um it's its set point is is I am engaging with novelty I'm engaging with what I don't know I'm engaging with what is uncertain um and if that sounds um paradoxical it's not it's because uncertainty is dangerous uh you know so so if if you have the opportunity to engage proactively with uncertainty now then you'll have less uncertainty later so what you're doing is actually building a better predictive model of the world you're you're engaging with the world and and better understanding how it works which in the long run uh enables you to better survive um in that world so the the the the thing that you just spoke of when you said don't we always want to keep on that's fundamentally driven by this it's not only seeking but it's fundamentally driven by the seeking uh the the the seeking homeostat um and then the third thing I wanted to say is that from all of this um voluntary behavior of us engaging with the with the world uh uh trying to uh find how do I get into back to where I need to be uh we learn from that experience the conscious experience is the feeling your way through the problem then you lay down new predictions based on what what you learned from that experience and there um the the the biological ideal is to have predictions which are so reliable that you don't have to feel your way through the problem again so that's again a trend towards zombom that in fact what we want is to have predictions that are so reliable uh that we that we don't um they don't need reconsolidation uh and so this is what we call non declarative memory you know these these these things that we just know how to do like how to drive a car you don't think what you know how does this work you just know how to do it how to ride a bike like how to walk you know um how to play the piano you know whatever these these these skills that we acquire uh they become automatized and that is actually a good thing so again in those situations I mean like I don't know if you play any musical instrument or like if you play tennis thinking about your stroke is a very bad thing to do you know it must just flow and so um in that sense too although we don't feel it that way way actually uh unconscious automatic processes are better than conscious deliberative ones when it comes to and learning from experience about how to

meet how to meet your needs okay now it's um so let's say so you had this hypothesis you've done the studies um you've already uncovered quite a lot U there's obviously a lot more to come but to you've already touched briefly on this but I think let's go into a bit more because there's the goal old standard I've wrote on in neuropsychology where you guys discover okay REM sleep versus Dream Within the brain stem when there's brain stem lesions and cortical lesions there's clear differences in REM sleep and dream sleep do you want to run through those and exactly what those lesions are yeah I I actually already have um said this so so let me just be a little bit more precise um the the the mesocortical molic dopamine system starts in the ventral tegmental area uh which is sort of at the transition between midbrain and dianlin uh IT projects upwards to the nucleus accumbent which is in the the ventral strium the underneath part of the basil ganglia that's the sort of main destination but it also projects uh beyond that uh up to frontal cortex and also down to amygdala so it's this it's the a very big part of the medial for brand bundle uh damage damage um anywhere in that circuit from the ventral segmental area upwards um certainly up to and including the nucleus cumbent uh but we've also had patients where it's anterior to the nuclear cir cumbent they they stop dreaming um the other area as I said earlier it's posterior cortex um it is uh in particular um that there two sites where we see cessation of dreaming and I say two sites it's really a continuous site but it goes from the the the the inferior parietal lobule down to the convexity of the temporal lobe but then also around into the medial surface along the occi temporal uh the medial surface so it's basically um it's basically a posterior Association cortex predominantly visual but not exclusive but by no means exclusively visual so lesions there those patients stop dreaming but they tend to recover they tend to recover um in one study that I did I I followed up 44 patients for a year uh and only two of them were still not dreaming but the vast majority recover within six months in fact and one of the reasons it's difficult to do research uh of the kind I was describing to you earlier when I said we were taking patients with the posterior lesions who could not dream for that reason and but the the the that dopamine system was was working fine um we wanted to see whether they will whether their sleep will be disturbed which as I say it was but many of our research participants recovered in the middle of while we were studying them because dreaming Tes tend it's a very robust phenomenon dreaming I think for the very reason that I said to you earlier that it's it's um if it has a sleep protective function then it's a very important thing you when you first started out in the field you you've done a lot you've already done so much pioneering work in in dreaming um what's brought you what's brought you back to this is it the link to Consciousness which we're obviously going to touch

on us soon but was it that link yes um in fact uh when I studied dreams uh which was I started my research in that area in the 1980s and uh I must tell you you know people don't realize uh those days uh Consciousness uh was not a respectable topic you know I mean can you believe it even in neuro Psychology you know which you would think you know psychology hello doesn't that have something to do with Consciousness it was no the only respectable topic uh as far as Consciousness was concerned was not the subjective EXP experience of anything it was it was the brain mechanisms of the of of of of sleep and wakefulness you know so the the how how uh the Sleep waking cycle uh is regulated and uh that the discovery of REM sleep was within that tradition you know the the these arousal states that occur during the sleeping part of the Sleep waking cycle and so I stud dreams uh I actually was studying brain mechanisms of sleep and dreaming and sleep was the intrusion of wakeful I mean dreams or the intrusion of wakefulness into sleep so wakefulness was a respectable topic and so you know I dressed up what I was doing as studying sleep and wakefulness dreams being wakefulness in in so um the the um the discovery of the brain mechanisms of dreaming at that time took me necessarily down into the basic arousal mechanisms of the brain stem but bearing in mind that it particularly took me to this one arousal mechanism which is the dopaminergic one this is um this is not just switching on and off the lights it's an emotional motivational mechanism and so that made me aware um how um made me interested in the fact that the brain these brain staral mechanisms are the brain mechanisms of emotion and and and that led me to to working with Yak PP because Yak PP was the was the only person who who said you know these these brain stem arousal mechanisms generate affect they they generate emotion they generate feeling and um then that so that was the second stage of my work you know from from from sleep and dreaming to emotion uh and then the realization which is the the the phase of my work that I'm in now that these mechanisms that generate feeling and affect and emotion are the very same mechanisms that generate Consciousness itself so that the foundational Elemental form of Consciousness is affective feeling and uh so it's you know it sounds as if I've been all over the park but actually it's really quite a straight line um it is I mean and and obviously in your paper you you lay this out perfectly you talk about okay this of course if this is effective feeling and if we're experiencing constantly within this dream state it must be something that's necessary during Consciousness well in conscious experience so this is a perfect place to actually look because you don't have to Consciousness can occur outside dreams have to include some sort of a conscious experience in a sense because it exactly yeah now that's a very interesting point what you just said there um so you

know one of the reasons that I um uh focused rather on the brain stem than cortex which also didn't win me any friends you know it's it's really to to to say in in neuro psychology that the cortex is not the organ of Consciousness is is really to say something that goes very very very much Against the Grain but but but a strong reason why I did that I mean there many reasons why I did that when I say reasons I mean you know there's much scientific evidence that led me to conclude that um but within that body of evidence is the fact that the cortex can do just about all of its cortical things without being conscious yes you know so I mean things that that only the cortex can do like recognize faces read with comprehension and so on you can read with comprehension and your behavior will be altered by what you've read without any conscious awareness of having seen anything at all so um most of the re when I say most I mean really just about all of the research uh on trying to to to identify what they call the neural correlative Consciousness um was was um using Vision as the cortical Vision because that's the conscious type of vision um as the as our model example but as I've just said to you um the cortex can do all of that visual gymnastics unconsciously so you know so so Vision even cortical vision is not an intrinsically conscious function so why do you think you're going to solve the hard problem of consciousness by studying something that is not actually necessarily conscious uh the reticular activating system and per aqueductal gray these brain stem structures uh that generate feeling um that their whole function is to generate Consciousness that's what that that's why it's called the reticular activating system it activates the cortex it's it's the cortex is only I mean this is not controversial the evidence is just like 100% there that the the cortex only is conscious when it is aroused by the reticular activating system magon and marutti showed in 1949 you sever as they did in cats the ascending fibers that you know the axons coming up from the reticular activating system to the cortex the cortex loses all Consciousness so the Consciousness is entirely supplied by the upper brain stem uh and it's only in recent years that we realized that that's not just a power our supply that Consciousness has a content and a quality it is affect it's feeling so we feel our way into our otherwise unconscious cognitions so um why that's important is because for the hard problem is that if you want to identify how and why does Consciousness arise you should be studying this basic in every sense the foundational form of of um of Consciousness a form that is called feeling which has to be felt in other words it necessarily is conscious so you know the way David Chas put it he said there's no cognitive function that the explaining of its function or mechanism will explain why it why you experience anything and and that he's right when it comes to vision and so on cognitive functions but you can't say that in relation to affect you

can't say there's no affective function you know that if you explain its mechanism you will explain why you experience it because how can you explain the mechanism of feeling without explaining why you feel it because you know that's the whole point of feelings is you feel them you haven't understood the mechanism of feeling if it doesn't explain why you experience it you know For Heaven's Sake so so I became interested in relation to the hard problem I became interested in how does the problem change uh if we look at this proper basic fundamental Elemental form of Consciousness called feeling now the interesting thing and this is why I was so interested in what you just said um when I've said you can't have a feeling without feeling it it is intrinsically conscious that's the function we should be looking at the same applies to dreams you can't have a dream without being conscious of it because otherwise it's not a dream you know the whole the the the the the the defining feature of dreaming in is that the unconsciousness of sleep is punctuated by this thing called the consciousness of a dream you know so uh and I don't think that that's I'm I'm not playing games here you know I'm not I'm not playing linguistic games or semantic games I think that there's a deep reason why that is the case it's because dreams are driven by affective functions you know those those affective functions are gener so affect is being generated during sleep in other words there's a problem that's Arisen for the organism you know it's in a state of uncertainty about something that it's now got to deal with and the dream is a way of dealing with that affective arousal in other words that deviation from where where we need to be it's it's a it's a biological problem that has arisen during sleep that needs a solution and the dream is doing the work of getting us of managing the feeling you know it's of of of in other words of solving the problem of of getting us back to homeostasis it's almost like it's so it makes so much sense that one should study dreams in order to understand Consciousness it's almost like why did we neglect this for so long oh well that's an easy question to answer it's because you know in fact it's the biggest problem of psychology is psychology is the science of the mind uh and what is mind if not something subjective the the the defining feature of of of mental life is that it is the subjective being of the organism it's it is intrinsically subjective I mean hence um you can ever you can you can only observe your own Consciousness you can't observe anybody else's Consciousness ever because they are objects you're you're a subject you can only experience your own subjectivity consciousness the mind subjectivity you know these these things are an embarrassment to science because science aspires to objectivity you know so how do you do science on something you know that you can only observe your own I know I I wrote one of your one of your lines in the paper is um scientists prefer objects over subjects

yes exactly and um now with dreams it's even worse because uh you're having a subjective experience while you're asleep uh you know so there there's no behavior that goes with this uh with this mental state the person uh the dreamer is asleep uh and so the only way you can get access to their dream uh is for them to wake up and remember the dream and then report it to you so it's a single witness uh retrospective report of a phenomenon that we know is notoriously forgettable uh and also inate most of the time it's also very hard to translate dream experience into words you know you you you know we all we all know that when you try to tell your partner or something you know what you dreamt last night you know as you're doing it it isn't really that wasn't really wasn't you know that doesn't quite convey what it was like so I think that um you know you said why didn't we start there uh in the first place I think that it's uh for these sorts of reasons it's also for these sorts of reasons that as I said to you a couple a good few minutes ago now I said that once the correlation between dreaming and REM sleep had been established we dropped the dreams and just studied the REM sleep but please note um what grave errors that can lead us into I mean we lost decades uh of research um effort because uh we we we we wanted to only study the object Ive side um of the of the Mind Body correlation um the the the um I think that let me say this the the brain is the only thing that we know of in the universe that has subjective experience um and uh surely this subject to this extraordinary quality uh it must be doing something you know and um if if that's the case you're never going to understand how the brain works unless you understand why unless you take account of its subjectivity you know the what the if the subjective I mean as we all know from our own lived lives you know you you know that what you do is caused in large part by how you feel you know and so if we leave that out of the equation we're never going to understand how that part of nature works there's also a lot of historical I mean you you you you've already said this but I mean the behaviorist movement obviously played a big role in this also the fact that dreams are often associated with Sigman Freud I mean the interpretation of Dreams is one of his biggest books and and a lot of people misinterpret a lot of Freud's work or because of the flaws within it people don't take it in a context and and then disregard a lot of it which is incorrect I mean one of your lines was consciousness of cortical representations contextualize affect and that context that context is so important yes look let's just go back to what you said there about the behaviorists because there's something that we must never forget um you know I told you uh I don't need to tell you the word psychology you know means science of the mind uh and the behaviorists excluded the psyche from psychology you know they said no we can't study the mental States because they're subjective we're only

going to study behaviors I mean how absurd is that to have psychology without the psyche is and and the the mistake there is to adjust the subject that you're studying to the methods that you have available you know you got to adjust the methods to the thing that you're studying um and embarrassing as subjectivity may be to science it's certainly exists in nature I mean the one thing that we can all be absolutely certain about is that we are experiencing as he told us it's the one thing you can be sure of is that you know you exist because you experience and um so you know you can't exclude it from science you got to find methods that uh that are suited to the study of subjectivity so that's what Freud did uh Freud's Freud I hastened to uh to to add for all of his mistakes with all of his faults you know Etc because you know I'm not one of those people who thinks that everything that Freud did was you know was correct by any means but I think that this thing that he did was correct was to say that the the the centerpiece of psychological science in in other words it's you have the psych psychological science has to has to be adapted to how do you study subjectivity and um so he didn't hesitate to study subjectivity the whole his method called free association it's basically just a sample of the stream of Consciousness uh uh um in a relatively controlled setting and um and he used the same method to study dreams they just getting the patient to or the research participant to say everything that comes to Consciousness in relation to the remembered elements of the dream and um in doing that he came to the not very obvious conclusion you know that dreams are motivated in fact Wishful states of Mind dreams don't look like that to to us most of them I mean what about nightmares for example most dreams just seem completely incomprehensible nonsense but Freud said no behind this you know and that's so there you have the weakness of his method because I mean how do you falsify that you can't falsify it to say behind every dream is a wish you know if it's uh behind the dream is the wish you know then then you can always find that wish if if you interpreting you know like he did that's that that's the method but now we have new methods so you know positron emission tomography you look at the dreaming brain and you see that that that motivating Wishful seeking mechanism switching on full blast and so you know I I I I my uh I don't know if you know this St but I I I became interested in Psycho analysis uh in large part because of my dream findings you know it's it was um it was you know sort of I was trained in an era where psychoanalysis was not part of the University it wasn't serious science at all you know so for me it was a really a big surprise to discover first of all that REM sleep is not what generates dreams secondly that what generates dreams is these emotional motivational circuits and in fact the most Wishful one that we've got you know is the one that made me take take a second look at Freud um I can only

imagine I don't even remember what your question was but that was my answer to it no it's fine but I can imagine because I remember once I mean you've said this in other podcasts as well but I remember you also told me that uh studying psychoanalysis was akin to people would describe it as almost being an astronomer studying astrology so ex there's so much backtracking you're already have to do and then in order to defend someone like Freud probably makes it even harder at that point um because people have so many negative connotations to it nowadays yes yeah that's right well think Sor sorry no I think that um if we can if we can re um introduce into psychological science and we are in our era we are you know reintroduce the subjectivity in the full sense of the word in other words not this function called Consciousness but rather you know the the actual experiential lived life the phenomenologic the the the qualitative stuff that is the proper subject matter of psychology um then you know we've we we we've we've rein re we've got back to where Freud started from and then we can go Way Beyond where Freud ended yeah know there is a field called phenomenological Psychopathology which a lot of people are trying to bring into Psychiatry and psychopathology just to assist with that lived experience aspect of it all but but anyway Mark let's go let's move on to the to to aect coming as I said you that quote of yours consciousness of cortical representations contextualize affect um then let's discuss those three features of affect first and move on from there I mean you describe them as veilance qualia and action when to go into detail with that before we move on yeah well the the thing that distinguishes affects from other mental States is the ve first and foremost uh uh affects are intrinsically Veen in other words they have an intrinsic goodness or Badness about them um and importantly that goodness or Badness is goodness and Badness to me for me it's subjective so it's it's um they are vean subjective States objects don't have intrinsic veence it's it's to it's it's this it's only this subject objective state that has this intrinsic veence this subjective State called affect um but pleasure and unpleasure veilance is a continuous variable you know you you start at this end there's unpleasure and you carry on you know through neutral to to pleasure it's a it's a you can quantify it on a single scale it's it can be reduced to to a common denominator um but uh affects um are are are not they are categorical variables in the sense that it's not just I feel good or I feel bad it's that I feel bad about what you know so the So Hunger is a particular need um which is different from thirst which is another need which is different from sleepiness which is another need um they all of them unpleasant uh but the they have to be categorically distinct from each other um because each of those needs has to be met in that in its own right you can't you can't say I've got eight out of 10 of um of sleepiness and

four out of 10 of thirst therefore I've got 12 out of 20 of total need um or total unpleasure um so therefore all I'm going to do is sleep you you I'm reducing the total number you you'll die if you don't you don't drink also so there this is a very important point you know I I it's such a simple point that I think that we can easily just sort of like skim over it but I really want to emphasize it that affects therefore must be qualitatively distinct from each other because they have to be treated as categorical variables it's it's it's there's no there's no alternative you have to you have to treat treat each need in its own right and so the subjective experience of the veilance has to be veilance in relation to this one in relation to this one in relation to this one and why that's important is because that means that because remember categorical variables are qualitatively distinguished from each other eight out of 10 of sleep is not the same thing as eight out of 10 of of thirst you know they are they're qualitatively different categorical variables work like that why that's so important is because there you have the necessary basis for qualia qualia have to exist in the mind in the sense that you have these subjective states which are intrinsically valanced and they also are intrinsically and I think that that's really important in terms again of the hard problem of Consciousness why do we have these qualitative States well there's one very good reason why we have them the third aspect that you mentioned um which is the behavior it's acting uh on those States uh because you know they are these are these are not nice to have you know these are these are needs these are phenotypic needs if you don't meet those needs you die I mean that's how homeostasis works so we have inbuilt behaviors um that they reflexes or instincts that go with each of these different um um phenotypic needs you know like um foraging behavior and and sexual behavior and and and and and and and sleep behavior and and you know Etc um but those are we're born with those things they're far too stereotyped they're not they're not even nearly uh enough to be able to enable us to meet our needs in the great unpredictable variety of situations that we find ourselves in including many novel situations and so we have to supplement uh these uh these behaviors with um through learning from experience uh and and so we have in addition to these innate predictions which we call reflexes and instincts we have then the full uh gamut of all of the different things that we do and and I just want to add this last point please note that everything we do ultimately if you look at it biologically is in order to meet our needs in order to stay alive in order to arrive and reproduce to take us back to that thing about the purpose of life that we were speaking about at the beginning of our conversation so continue Mark sorry I was no no that's that's that's what I wanted to say so that's what that's what a effects are it's it's intriguing because um I mean at some point in your

paper you eventually bring up bring up the knowledge argument to to sort of solidify this aspect um I was sharing by email with Frank Jackson about that um try trying to get him on the show but he says he's taking a break from social media because I think that's one of the best arguments to show someone an example of I mean that that pure essence of feeling um when Mike's episode came out had people going at it in the comment section you have people discussing how because Mike at some point had said that the only way to truly know to truly get a conscious a future where we can all integrate and corporate better is if we slowly learn what each other are experiencing more and more and the only true way to know what someone else is experiencing is to actually experience what they're experiencing because otherwise there's no other way it's it's pretty much an impossible reality at that point um and then people were bringing up the knowledge argument and saying okay well if they have all the theory why is that impossible do you want to touch on that and give them an answer the the oh well just in case there's people uh watching or listening who who are not familiar with the knowledge argument uh it it is in simplified form uh this is my simplification of it that there's there's a visual neuroscientist named Mary and she knows everything that there is to know about the mechanisms of vision you know the physical physiological processes whereby visual information is processed but Mary's blind that's my simplified part of it and so she doesn't know what it is like to see she knows everything that's the knowledge she's got all the knowledge about uh how visual information is processed in the brain but she knows nothing about what it is like to see the actual qualitative phenomenological experience of vision and so Jackson's argument those days because he's changed his mind by the way but in 1982 when he when he published that he said that shows that the actual qualitative phenomenological experience of seeing is not accounted for by the causal physical mechanism it it exists in some other plane um nothing about her knowledge of the physical mechanisms of vision prepared Mary for what it is like to see e even gave her any reason to predict that there would be something it is like to see it seems therefore that the the the the qualitative subjective experience of seeing isn't necessary it doesn't do anything it's just you know and where does it fit into the mechanism it doesn't that was the knowledge argument so my response to that was that's because as I've said previously in this conversation because vision is not an intrinsically conscious function you can understand everything about how visual information processing works because without reference to what it is like to see because visual information processing can work without you having any experiential Vision um I mean your phone can see you the camera in your phone sees a face recognizes the face takes a photograph of a face remembers

their for that face all of that stuff but there's nothing it's like to be a camera or to be a phone and it does it even better cu if you are if you click take remove the nose the phone knows exactly which parts to remove so it sees even better than us yeah so all of this very complicated visual information processing can go on without any experience now as I said earlier that does not apply to another function so that's the function of vision but the function of feeling you cannot explain the function of feeling without knowing what it is like to feel without explaining why it feels like something if if Mary was an affective neuroscientist actually first of all if Mary was an affective neuroscientist who'd never experienced affect she would in fact be dead because her brain stem mechanisms that keep her alive you know would would not be working and in in in a manner of speaking that's also an important point it's that you know this is not this is not fluff feeling is is is has massive causal consequences but patients in coma you know they die unless we keep them alive you you can't live without without being conscious and um you know in other words the the Consciousness is doing something bloody important um so so that was my argument is that had we started with affect rather than with vision the the the the the knowledge problem would never ever risen and and and the hard problem which which comes 30 years later Chalmers Builds on Jackson's knowledge argument in fact he it's a very pivotal to Chalmers as whole view that you know you can explain the function without explaining why there's an experience attached to the the the um execution of that function uh that's pretty much the essence of the hard problem also I made a not oh sorry I made a Nota saying unconscious feeling is an oxymoron you also like yeah that's a very nice one yeah but but please remember T what I'm saying is that it's not only that it's an oxymoron it is an oxymoron but it's an oxymoron for a very good reason yes you know it's and when I say that the the that feeling is where we should feeling is the elemental form of Consciousness and we should study that if we want to understand its mechanism how it arises why it arises um then let us not forget that the part of the brain that generates feelings is the part of the brain that generates Consciousness it's like this is intrinsically Consciousness generating tissue so it's not just a word you know that feeling must be felt it is that the what this part of the brain is doing is generating states of of Consciousness it's the whole the whole reason that the that the arousal activating systems are there I have to say because when I first when I first started reading the the your work on affect and the way you were approaching Consciousness I remember also initially thinking this might this is a bit out there so when you say like people people might think it's crazy and stuff I I can get where you're coming from but the more you read it the more you see the evidence the more you talk about it it it becomes

very it almost becomes very obvious it's almost like you're highlighting a problem that we've been ignoring for quite some time how do you feel about it do you feel like it's it's it's about time people took this approach or or do you think it's gaining some Traction in the way people perceive the problem um well I'll tell you since we started our conversation this conversation now um You you we were talking about my recent trip to Boston and um other speakers at that meeting uh there's Mike Levan Chris Fields um Carl Friston um uh um what's his name um Richard I forget his surname Watson yes you know these are apologies Richard Watson if you're watching these are um really really outstanding Minds you know and so I I my paper I said to them look uh you know I wanted to because the conference was mostly about biological versus artificial uh minds and um I said you know I'm doing work in that area and I really want to get there to talk to you guys about this but I want to spend at least the first half of my talk just setting out my stall for why I believe that the we're looking in the wrong place it's not cortical it's a it's you know it's a brain stem function and it's an affective function and so on and um I set out all the evidence for them I said the reason I'm doing this is because to my great surprise you know many people still you know don't believe this and I just I just can't understand it what am I doing wrong that you know that I'm can't you see the evidence and so so at that conference in in because you're asking me did it gain traction all of those guys who I just mentioned they all were 100% onite you know so yeah I I think well gosh you know so that's very very gratifying but I want to while I'm busy saying immodest things um I want to go back a little bit earlier than this Boston Congress I mean Symposium uh you know Tom Nagel the other day the Consciousness conference he he gave the keynote paper um and he Tom Nagel who wrote the 1974 paper what is it like to be about spoke about everything that's happened since then in Consciousness studies and mean it's really like the kind of the the great grandfather of of of of the modern philosophy of of of of Consciousness um he's in his talk about 2third of the way through it I mean I couldn't believe my ears I'm sitting sitting there listening to Tom Nagel It's a Wonderful treat you know G giving us this great sweeping sort of description of where he started and where things have gone since then and where things are now and where how he sees the future how does he see the future well he says I think this guy called Mark somms is doing the right thing he's he's pointed out that feelings are intrinsically conscious States and I think that's right you know and I think that that's the way forward I thought wow I've arrived you know Tom in that he name check two people one was Mark SS the other one was bar Spinoza wow that's I mean that's a lot uh that's that's that's one of the best name drops you can get from I mean if you think of Tom Nel it's he's he's like the

father of modern philosophy of mind at this point the Tom Nagel's probably the most quoted philosopher and I think David Chas with his heart problem is probably follows right after but so that that's pretty epic yeah yeah I mean I I was really I mean not only because remember you asked me is it gaining traction I mean I wasn't only gratified and I was deeply gratified but I was to tell you the truth I was frankly surprised you know I really I really I would never have thought that Tom Nagel would say no actually I think we are making progress toward a physical theory of of of experience you know that's that's that's really news well well I just want to say congrats if that comes out of Tom's mouth it must have been surreal in the moment yeah it was it really was I mean but even from my own experience Mark when I speak to when I spoke to Carl FR when I spoke to these other guys about your work it there's clearly a huge respect for it there's these guys clearly understand it they know what you're talking about and they think the approach is is is a good approach it's not it's not one that they're discounting when they when they discuss these matters yes that's that's um no I I I really I I think that it's true to say that it is gaining traction and but it's also true to say that it wasn't the case initially you know it's um because I also at the time when I I remember when I first started the podcast they were some people who who think cuz I think they tend to associate your Viewpoint at this point with just an ancient brain um and there's almost a negative connotation to saying those words as if you're you're talking about something ancient in itself well yeah no and I I understand where that comes from you know it's because in fact I I must say I blame Paul mlan for that you know this whole thing about the Triune brain that you know that there's this kind of cartoon idea that there's a Reptilian Brain you know sort of trapped inside of our inside of our brains now so I want to make clear that when I say ancient what I mean is phog genetically ancient in the sense that because we have because all vertebrates have the architecture of the upper brain stem that I'm talking about because we share with all vertebrates a reticular activating system and a per aqueductal gray that means we have a common ancestor that had that structure the common ancestor of all vertebrates is 550 million years ago so I'm talking about that guy the 550 million years ago that's what's ancient I'm saying that this structure if evolved and what that what that means is that the dawn of Consciousness if I'm right that this structure is what's generating the the brw feeling you know the basic the most elementary form of Consciousness the dawn of Consciousness was at least 500 at least 550 million years ago that's what I mean by ancient I think there's there's there's a lot of misinterpretations but because of the fact that you're going to these conferences you have people like Tom Nagel now obviously supporting you hopefully people understand it

more or give it more time in order to gain that understanding because I think that's a fundamental approach is to actually read your work read your papers and then draw a better conclusion yes you I suppose the you know the best thing that can happen is that people read your work and they're persuaded by it um the the second the second best or middle thing out of three that can happen is they read your work and they disagree with you and they reject it the worst thing that can happen is that they just ignore it completely so you know even even when you initially get um negative uh know push back at least it's not just completely overlooked yes you know it's true it's sometimes naming a theory well gives it that that that that negative push back you need so like sometimes saying things like the ancient brain is like Keith Frankish or denn saying it's an illusion it it helps with the name that illusionism gives it that push that it almost needs time while you're on that subject I know we we're nearly out of time but I want to I want to say something else about feeling um which is that um unlike perception it cannot be an illusion uh you you you can't say uh you you are currently having the illusion that you're in pain uh you you you just are in pain uh you you can't say that um you having the illusion that you're sad you might say you have no good reason to be sad but still you're sad and I think that that's also important you know that when we when we speak about um about um what do we mean by Consciousness as an illusion or I I I think it's not this doesn't have exactly the same meaning when you speak about an illusion that the brain constructs its own sort out of its own ingredients it constructs uh the illusion of a three-dimensional world out there with colors uh that actually the colors are a product of your brain rather than a rather than a property of the world um feeling it's a because it's about yourself you know it just is what it is it's it's you can't have illusory feelings and it's yeah because I was just about to say it's almost like the argument of illusionism can't work against your view of Consciousness because it's fundamentally talking about different things because the the illusion has come at it from a very much cortical perspective in the first place so yes now I realize while we're speaking about this that I didn't address a point you made earlier and it suddenly becomes relevant again now you said what did I mean when I say that that that perception contextualizes affect um and this is exactly the right place to come back to that so how it works is that you have the feeling that's just that is just the consciousness then it's a matter of what that what what is causing that feeling and what can I do about that feeling so I I like to use this um termin this um this wording but of course I'm translating something into words that is not verbal it is first of all the creature feels like this it's just raw feeling I feel like this and then secondly it's a matter of I feel like this about that and so the and so the affect is as

it were applied to the context within which it's with within which it's occurring so this oh so it's these things that are making me feel like this and it's there that I can do something about this feeling and so this is what I mean when I say we feel our way into our cognitions and we feel our way into our perceptions um yeah like that approach yeah so so the the you said it's good to have a catch phrase catchy phrase you said it it's it's good to have a controversial one but the I was somebody recently wrote a a review of different theories of Consciousness and they wanted me to tell them what's my theory called you know so I said I'm calling it The felt uncertainty it's the felt uncertainty Theory and um I think that that's another way in which you can put what I've just said now it's like I'm you know it's it's like the common currency is uncertainty it's like I feel like this uh in other words I'm I'm I'm I'm I'm not in a state of certainty otherwise you wouldn't be feeling the feeling always remember is a deviation from homeostasis and then it's a matter of you know so you apply that uncertainty to your to your actions um feeling your way through is this working is that working is this working is that working you know so it's a it's a matter of um affects uh sort of um you palpate your confidence in your current Action Program on the basis of whether or not it is moving the needle in the right direction effectively so so it's it's um I like to put it like in fact it was Carl who put it like this he said that um if things are turning out as expected on the basis of my actions if things are turning out as expected that's good and if uncertainty prevails bad okay yeah so it sort of it allows you to gauge the environment and interact with it in a very successful Manner and to maintain homeostatic uh in order to maintain homeostasis sorry order to maintain homeostasis the felt will you call it the felt uncertainty principle yeah that's it's quite a nice I was thinking about it as well before coming into this conversation how call has one of his papers he calls it I I am therefore I think Kevin Mitchell has I feel therefore I am I mean I move therefore I am and you could make it I feel therefore I am is another one yeah just to just to add that to the to the lecture Series in the future um Mark before we finish one last thing about this um the conference was about AI anything you want to add about feeling and artificial intelligence and the implications for the future yes uh I I am working um with a team of um physicists and and uh roboticists and and computer scientists on an um an AR arcial Consciousness we've been at it for about two years um we we've got some good funding from the opers by the way um and um we we're now going into our second phase and um you know I I I'm really very optimistic about what we're doing there um we're trying because we believe that we've isolated at least in Broad outline the causal mechanism you know how how feelings are generated how and why they generated if you've if you've done that uh if you

really have done that then you should be able to engineer it in fact it was Richard feynman who said right at the end of his life it was found on his on his Blackboard if I can't create it I don't understand it and so you know this is what we're trying to do we're saying well if this really is the mechanism then we should be able to engineer artificial feelings remember we're not creating a living thing we're creating a an artificial agent that has exactly the mechanisms I was describing to you earlier about different needs and having to balance those needs and they are intrinsically veilance to and for that agent and they are categorically distinct and etc etc and then this should be able to display the kinds of behaviors that I was describing to you the where you can feel your way through a problem come up with novel Solutions it should display voluntary behaviors and it should be able able to survive in unpredicted environments and so on so these are the sorts of things we're doing and I really do believe it's a matter not of whether but when uh we'll we'll succeed with that it's another story about how do you persuade your colleagues you know the the problem of other minds and all of that is not a small thing uh what what we are going to be doing is having a symposium with interest parties where we must determine in advance what will be acceptable criteria you know that you can't say there's no way that you will ever persuade me at all you know you have to say well what will persuade you yes and um and then the ethics of that you know we we must always always keep very clear in our minds that the the ethical there's two sets of ethical problems the one is you know the one that people most frequently refer to is but you you're going to create these super intelligent uh creatures that are trying to survive and they're going to compete with us and you know that's bad for us but we must also remember that if you create a an artificial Consciousness if it feels uh then you know it can suffer and uh it you have to be very careful about what are you what are you producing here you know and it's all of the problems that we have in animal research uh become relevant then to this sort of research too Mark I'm sorry I don't know how much time you have left but I'm so curious to know how does one even approach this how do you even start building this sort of artificial complexity well um you start with the formalisms that uh was Carl and I published a paper in 2018 you know he published that brilliant paper in 2013 called Life as we know it and there he reduced two formalisms uh the mechanisms of fundamentally it was the mechanisms of homeostasis that's what because the whole mechanism of self-organization is what underwrites the mechanism of homeostasis when he says Life as we know it I mean homeostasis is the mechanism of Life as we know it it's it's what distinguishes living things um so what was I saying yeah so so that the formalisms that he developed to explain uh homeostasis we extended those formalisms to

explain felt homeostasis in our 2018 paper and uh then you know I I got this team together and we're using those to develop AI that formalism to develop algorithms to create artificial agents that have homeostatic needs in order for them to continue to survive it's just a simple mechanical unpacking of this thing it's creating the exactly those Dynamics uh in an artificial agent and um and creating an environment within which uh we can have the sorts of um dynamics that we need in other words changing so the the agent has to meet various needs of its own like power supply and all of that and and uh then the resources where they are we move them around once it's learned okay this is this is how if I do this I can survive then we change it you know and then we and then we created situations where it's not just a matter of exploiting resources but also there's a hill that it can go up and have a look around we want to see does it realize that if I go up there I see the I see so that's sort of roughly where we are now we we we're we're um we've got a a long way to go but a major part of the next phase that we're going to do is to have multiple agents so they in where they have to interact with each other and they have to they have to um make inferences about each other's mental processes um in in in is uh you know as part of the same thing oh that's super exciting I'm sorry about the interruption but I'm going to have to go it was really great talking to you again tev Mark it's been a pleasure always and uh thank you so much for I know you don't like to call it a round two it sounds like a boxing match but hopefully there's a round and many to go as you continue on this amazing journey thanks again mark thank you so much thanks for your interest and your time always always pleasure cheers